Use models of energy flow between the geosphere, biosphere, hydrosphere and atmosphere to explain patterns of global climate change

Learning intention

- use models of energy flow between the geosphere, biosphere, hydrosphere and atmosphere to explain patterns of global climate change.

Teach and model

- examining the role of radiation from the sun and how its interactions with the atmosphere.
- investigating how quantum computers enhance modelling of complex weather and climate systems.
- predicting changes to the Earth system and identifying strategies designed to reduce climate change or mitigate its effects.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.